Uniform Vacuum Weights and Harnack Control
Target 55 of the H1 Downstream Programme

Yang–Mills representational dynamics research programme

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Abstract

Finite Perron positivity does not control the inverse vacuum weight uniformly. We identify a sharp elementary sufficient hypothesis: an entrywise positive transfer amplitude with uniform lower and upper bounds. It yields an explicit Harnack ratio and root-moment bound. A positive two-sector family then shows exponential loss when the lower bound degenerates. This isolates the physical Yang–Mills obligation as a uniform comparison principle rather than a finite eigenvector computation.

1 Finite Harnack theorem

Let T be an $N \times N$ strictly positive matrix and let $T\psi = \lambda_0 \psi$ with $\psi > 0$.

Theorem 1 (Entrywise Harnack bound)**.** *If $0 < m \le T_{ij} \le M < \infty$ for every i, j, then*

$$\frac{\max_i \psi_i}{\min_i \psi_i} \le \frac{M}{m}. \tag{1}$$

Proof. For any rows i, k,

$$\frac{\psi_i}{\psi_k} = \frac{\sum_j T_{ij}\psi_j}{\sum_j T_{kj}\psi_j} \le \frac{M \sum_j \psi_j}{m \sum_j \psi_j} = \frac{M}{m}.$$

Interchange i, k and maximize. □

Normalize the forward cost by $V_i = (\max_j \psi_j)/\psi_i$.

Corollary 1 (Uniform root moment)**.** *$1 \le V_i \le M/m$, and hence $\nu(V) \le M/m$ for every probability root law ν.*

The associated Doob kernel

$$P_{ij} = \frac{T_{ij}\psi_j}{\lambda_0 \psi_i}$$

is stochastic. Thus the same hypothesis controls both the normalization of the dynamics and the forward inverse-amplitude cost.

2 Finite calibration and hostile family

For the displayed three-state positive matrix in the executable certificate, the actual amplitude ratio is 1.1222312626, while the entrywise theorem gives the conservative bound 5.

Now set $\epsilon_L = e^{-L}$ and

$$T_L = \begin{pmatrix} 1 & \epsilon_L \\ \epsilon_L & 2\epsilon_L^2 \end{pmatrix}.$$

Every T_L is strictly positive and positive definite for sufficiently large L, yet its suppressed-sector inverse weight grows like e^L. At $L = 20$ the actual ratio and hostile root moment are approximately 4.851651954×10^8, while the entrywise bound has deteriorated to $1.176926334 \times 10^{17}$.

Warning 1. *Finite-volume Krein–Rutman or Perron–Frobenius positivity cannot be promoted to a cutoff-uniform root moment. A global entrywise minorization may itself be too strong for Yang–Mills; a chart-chain, sector-aware Harnack inequality or a direct weighted moment estimate would suffice, but must be proved uniformly.*

3 Partition-weight diagnostic

Given boundary partition weights Z_i, the normalized vector Z_i^{-1} can be compared with ψ_i^{-1} by the oscillation of their pointwise ratio. This is a diagnostic only: equality requires a theorem identifying the temporal transfer amplitude with the retained conditional disintegration.

4 Audit

The Harnack and root-moment implications are C2 finite theorems. No cutoff-uniform physical comparison, basin entry, continuum construction, or gap follows. Ten hostile tests verify the eigenvector, Doob, Harnack, root-moment, mismatch, and nonuniformity claims.